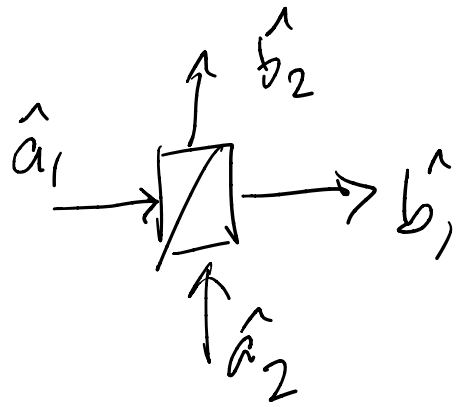
10/14/2026

Kok & Lovett



$$\begin{pmatrix} \hat{b}_1 \\ \hat{b}_2 \end{pmatrix} = \begin{pmatrix} \cos\theta & -ie^{i\phi}\sin\theta \\ -ie^{-i\phi} & \cos\theta \end{pmatrix} \begin{pmatrix} \hat{a}_1 \\ \hat{a}_2 \end{pmatrix}$$

θ - describes the reflectivity

ϕ - relative phase between transmitted & reflected ports

If $\phi = 0$, $\theta = \frac{\pi}{4}$ $\quad U_{BS} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}$

symmetric BS

$-\pi/2$ relative phase between reflection & transmission

$\phi = \frac{\pi}{2}$ $\theta = \frac{\pi}{4}$ $U_R = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$

$$U_R = D^\dagger U_C D \qquad D = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

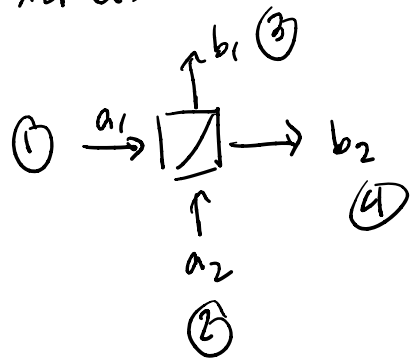
Both U_C & U_R predict the HOM interference

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} t & r \\ r' & t' \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

$|t|^2 + |r|^2 = 1$ → energy conservation

$r' = -r^*$ $t' = t^*$ → unitarity

Exercise: HOM interference



$|\psi_{in}\rangle = |1\rangle_1 |1\rangle_2 = a_1^\dagger a_2^\dagger |0\rangle$

$a_1^\dagger \to t b_1^\dagger - ir b_2^\dagger$

$a_2^\dagger \to t b_2^\dagger - ir b_1^\dagger$

$$|\psi_{out}\rangle = (t b_1^\dagger - ir b_2^\dagger)(t b_2^\dagger - ir b_1^\dagger)|0\rangle$$

$$= (t^2 - r^2)|1\rangle_3 |1\rangle_4 - irt\, b_2^\dagger b_2^\dagger |0\rangle - irt\, b_1^\dagger b_1^\dagger |0\rangle$$

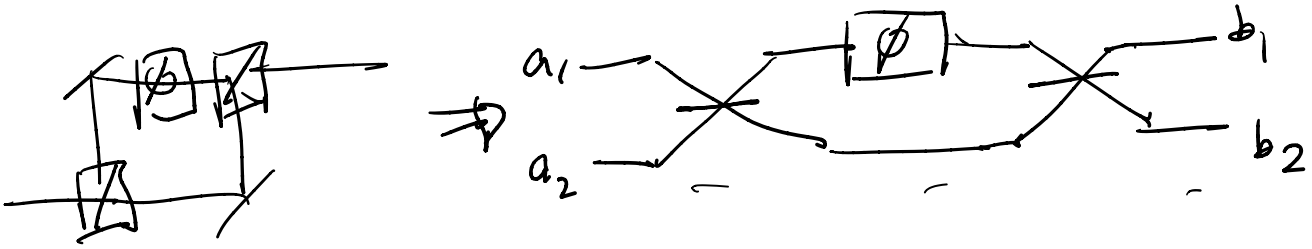
$(b_2^\dagger)^2 |0\rangle = \sqrt{2}\,|2\rangle_3$

$(b_1^\dagger)^2 |0\rangle = \sqrt{2}\,|2\rangle_4$

$$|\psi_{out}\rangle = (t^2 - r^2)|1\rangle_3 |1\rangle_4 - \sqrt{2}\, irt\, |2\rangle_3 - \sqrt{2}\, irt\, |2\rangle_4$$

Phase gate : $P(\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}$

e.g. BS + phase gate = Mach-Zender interferometer



$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

$$P_1 = \cos^2 \frac{\phi}{2} \qquad P_2 = \sin^2 \frac{\phi}{2}$$

$$U_{MZI} = U_1^{+} \, P(\phi) \, U_1$$